Machine Learning-Enhanced Bus Identification and Route Finder

The *Machine Learning-Enhanced Bus Identification and Route Finder* system is designed to modernize and optimize university transportation management by integrating intelligent automation with user-friendly digital solutions. Traditional bus management systems rely heavily on manual processes, which often lead to inefficiencies, delays, and lack of real-time information access. This project addresses these limitations by introducing a smart, technology-driven system that improves both operational efficiency and user experience.

The core objective of the system is to provide an automated platform that enables users—students, faculty, and staff—to easily access accurate bus information, schedules, and routes. A significant innovation in this system is the integration of machine learning and image processing techniques for bus identification. By analyzing images, the system can detect buses and extract key details such as license plate numbers, ensuring accurate tracking and minimizing human error. This feature enhances reliability and introduces a level of automation not commonly found in traditional transport systems.

In addition to intelligent identification, the system incorporates a robust timetable management module. Administrators can efficiently add, modify, and manage bus schedules, while users can view updated timetables in real time. The system also includes a route-finding feature that allows users to search for buses based on routes or numbers, significantly improving accessibility and convenience.

Another important component of the system is its user interaction capabilities. Users can create accounts, receive notifications about delays or route changes, mark favorite routes, and provide feedback. These features ensure active user engagement and continuous system improvement. The feedback mechanism also enables administrators to monitor service quality and address issues effectively.

From a technical perspective, the system is built using modern tools and technologies, including Flutter for cross-platform application development and Python for implementing machine learning algorithms. The system also utilizes databases for efficient data storage and retrieval, ensuring scalability and performance.

Overall, this project aims to transform the conventional bus management process into a smart, automated, and user-centric system. By combining machine learning with efficient software design, it enhances transportation reliability, reduces manual workload, and improves the overall commuting experience. The proposed solution not only meets current transportation needs but also provides a scalable foundation for future enhancements such as real-time tracking and advanced analytics.